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Department of Biological Sciences

BIO204 Advanced Genetics - Lab Report 1

Name of Practical: Genotyping and Population Genetic Analysis

Group Number: 35 (Thursday)

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Population Genetic Analysis Reveals Hardy–Weinberg Equilibrium at TPA-25 and Polymorphic STR Alleles at the VWA Microsatellite Locus

1. Abstract

Two commonly used population genetic markers, the TPA-25 Alu insertion and the VWA microsatellite locus, were analyzed using PCR and gel-electrophoresis based genotyping of genomic DNA extracted from cheek cells. At the TPA-25 locus, genotypes were determined by the presence (400 bp) or absence (100 bp) of the Alu insertion. Among 191 valid samples, the insertion allele frequency was p = 0.542, and genotype frequencies did not deviate significantly from Hardy–Weinberg equilibrium ($\chi^2 = 0.722$, $p = 0.697$). For the VWA locus, fragment migration distances measured from gel images were converted to fragment sizes using a log-linear regression calibration based on a DNA ladder. Seven allele classes were identified between 138 and 162 bp in 4-bp repeats, with 154 bp representing the most frequent allele. Together, these results confirm the insertion polymorphism of the Alu locus and the size-based allelic variation characteristic of STR microsatellite loci, demonstrating how different classes of genetic markers reveal patterns of human genetic variation and population evolution.

2. Introduction

2.1 Population Genetic Analysis and the Hardy–Weinberg Principle

Genotyping and population genetic analysis examine how genetic variation is distributed within populations and how allele frequencies change across generations.

A central theoretical framework for analyzing such variation is the Hardy–Weinberg (HW) principle, which states that allele and genotype frequencies remain constant from generation to generation in the absence of evolutionary forces (1,2). The model assumes no selection, mutation, or gene flow, random mating, and an effectively infinite population size. Under these conditions, the expected genotype frequencies for two alleles with frequencies p and q are p^2, $2pq$, and q^2, where $p + q = 1$. Deviations from HW expectations, therefore, indicate that evolutionary processes may influence such a locus (3).

2.2 Genetic Markers and Evolutionary Evidence

Genetic markers enable alleles at specific loci to be distinguished and their frequencies quantified across individuals, providing a basis for analyzing genetic diversity and population structure (4).

Different classes of markers evolve at different rates, ranging from thousands to millions of years. Consequently, particular markers are selected according to their evolutionary properties for applications such as forensic identification and parentage analysis (5).

2.3 SINE, Alu Elements, and TPA-25

Alu elements are primate-specific short interspersed nuclear elements (SINEs) derived from transposons. They are approximately 280–300 bp long and originated from the 7SL RNA gene, which encodes the RNA component of the signal recognition particle involved in protein export (6). More than one million copies of Alu sequences are present in the human genome, accounting for nearly 11% of total genomic DNA (6).

Although most Alu insertions are fixed in the human genome, a subset remains polymorphic and contributes to human genetic diversity (6). One such polymorphic locus is TPA-25, a dimorphic Alu insertion located within an intron of the tissue plasminogen activator gene on chromosome 8 (7,8).

2.4 STR, Microsatellite, and VWA

Microsatellites are DNA sequences composed of short tandem repeats (STRs), typically 1–6 bp in length (4). They are widely distributed across eukaryotic genomes and display high levels of polymorphism due to variation in the number of repeat units. Most microsatellite variation arises from slipped-strand

mispairing during DNA replication, generating alleles that differ in length by multiples of the repeat motif (9). Owing to their high mutation rate and co-dominant inheritance, microsatellite loci have become widely used markers in population genetic analyses.

One commonly studied microsatellite marker is the VWA locus, located on chromosome 12p12–pter. This locus contains tetranucleotide repeat motifs of the form $TCTA(TCTG)_{(3–4)}(TCTA)_n$, producing fragments typically ranging from 138 to 162 bp, depending on the number of repeat units present (10).

2.5 Aim of the Study

This study examines human genetic variation using two distinct classes of molecular markers. The TPA-25 Alu insertion locus was genotyped to assess conformity with Hardy–Weinberg equilibrium and provide an assessment of insertion polymorphism. In parallel, allelic variation at the VWA microsatellite locus was analyzed to characterize the distribution pattern of STR alleles within the population. Together, these analyses experimentally illustrate how different marker systems can reveal patterns of genetic polymorphism in human populations.

3. Materials & Method

3.1 TPA-25 Genotyping

3.1.1 DNA Extraction

DNA samples were obtained from volunteers enrolled in the BIO204 practical class. All samples were anonymized before analysis.

Genomic DNA was extracted from cheek epithelial cells. Cells were collected by scraping the inner cheek with sterile mouth swabs and transferred into bijou tubes containing 1000 µL 0.9% NaCl. Tubes were vortexed for 15 s to release cells into suspension. The white cloudy suspension was transferred to a 1.5 mL microcentrifuge tube and centrifuged for 10 min at 11,000 rpm using Thermo Pico17 or Sigma 1-15 microcentrifuges to pellet the cells. After discarding the supernatant, the pellet was resuspended in 500 µL 10% Chelex resin, which binds metal ions that inhibit polymerase chain reaction (PCR). Pipette tips were slightly cut to facilitate the transfer of the Chelex suspension. Samples were incubated at 100 °C for 10 min with intermittent agitation to lyse cells and release DNA. Tubes were then cooled on ice for 1–2 min and briefly centrifuged (30 s) to pellet Chelex beads and debris. 200 µL of the supernatant containing genomic DNA was transferred to a clean 0.5 mL microcentrifuge tube and stored on ice until PCR amplification.

3.1.2 PCR Amplification

The TPA-25 locus was amplified using PCR in a 25 µL reaction volume (Table 1). PCR was performed in a thermal cycler under the following conditions: an initial denaturation at 94 °C for 2 min, followed by 30 amplification cycles consisting of 94 °C for 30 s (denaturation), 58 °C for 30 s (primer annealing), and 72 °C for 45 s (extension). The reactions were then held at 4 °C until further analysis.

3.1.3 Agarose Gel Electrophoresis

PCR products were separated using 2% agarose gel electrophoresis prepared in 1× TAE buffer. Agarose (2 g per 100 mL) was dissolved by heating and cooled to approximately 60 °C, after which GelRed (1:10,000) was added. The gel was poured into a casting tray with a comb and allowed to solidify for ~30 min before placement in an electrophoresis tank containing 1× TAE running buffer. PCR products were mixed with 6× loading buffer and loaded alongside a DNA molecular weight marker. Electrophoresis was conducted at 100 V for ~45 min. DNA bands were visualized using a gel imaging system, and digital images were recorded.

3.1.4 Genotype Determination and Population Genetic Analysis

Genotypes at the TPA-25 locus were determined from PCR fragment sizes observed on agarose gels. Fragments of approximately 100 bp indicated the absence of the Alu insertion, whereas fragments of approximately 400 bp indicated its presence. Individuals were classified as 100/100, 100/400, or

400/400 genotypes. Band classification was independently performed by three experimenters and subsequently cross-validated.

Allele frequencies were calculated from genotype counts. Expected genotype frequencies were estimated and compared with observed values using a chi-square test in R (version 4.4.3).

3.2 VWA Genotyping

3.2.1 Band Measurement and Calibration

To improve fragment resolution, polyacrylamide gel electrophoresis was used instead of coarse agarose gel.

Gel images containing VWA PCR products were analyzed using ImageJ (version 1.54p) to measure DNA band migration distances. Distances were recorded in pixels from the bottom of the well to the center of each band, and each band was measured repeatedly to obtain an average value.

Fragment sizes were estimated using a 100 bp DNA ladder as the calibration standard. Because DNA migration distance in gel electrophoresis is approximately proportional to the logarithm of fragment size (11), migration distances of ladder bands with known fragment sizes were measured and used to construct a regression model relating migration distance to $\log_{10}$(fragment size) in R.

3.2.2 Fragment Size Estimation

Fragment sizes of sample bands were predicted from the regression model using their measured migration distances. Estimated fragment lengths were then assigned to VWA allele classes ranging from 138 bp to 162 bp in 4 bp increments.

Reagent	***Volume (μL)***
DNA sample or H2O (blank)	*5.0*
10 X PCR Buffer (with dye)	*2.5*
TPA-25 (F, 10 mM, 20X)	*1.25*
TPA-25 (R, 10 mM, 20X)	*1.25*
dNTP mix (10 mM, 50X)	*0.5*
Q-solution (5X)	*5.0*
ddH2O	*8.5*
Taq polymerase	*1.0*
Total	*25*

Table 1. *Composition of the PCR reaction mixture used for amplification of the TPA-25 locus.*

4. Results & Analysis

4.1 TPA-25 Data Analysis

4.1.1 Genotype Recognition

PCR amplification of the TPA-25 locus produced fragments reflecting the presence or absence of the Alu insertion. Agarose gel electrophoresis resolved two characteristic fragment sizes: approximately 100 bp, indicating the absence of the insertion, and around 400 bp, indicating its presence (Figure 1A). Individuals displaying a single 100 bp band were classified as homozygous for the absence of the insertion (100/100), whereas those showing a single 400 bp band were classified as homozygous for the insertion (400/400). Samples containing both bands were identified as heterozygous (100/400). Representative examples of these genotype patterns were observed in the gel image from Group 35, showing two homozygotes for the insertion (400/400) and one homozygote for the absence of the insertion (100/100) (Figure 1B, Table 2).

$$p + q = 1 \quad (2)$$

4.1.2 Observed Genotype Distribution

Genotype records were collected from 217 volunteers. Among these samples, 26 gel images were excluded because bands were too faint for reliable scoring, showed additional bands outside the expected 100 bp or 400 bp positions, or contained apparent primer dimers, defined here as very small non-specific PCR products formed by primer–primer amplification and clearly distinguished from the 100 bp allele band. These samples were removed from subsequent analyses. After quality filtering, 191 valid individuals remained. Their genotypes are summarized in Table 3, which represents the observed genotype distribution at the TPA-25 locus in the analyzed population. Among the 191 validated individuals, 43 were classified as 100/100, 89 as 100/400, and 59 as 400/400.

4.1.3 Allele Frequencies and Expected Genotype Frequencies

Allele and genotype frequencies were calculated according to the Hardy–Weinberg principle.

Allele frequencies were derived from observed genotype counts using Equations (1) and (2). The frequencies of the presence and absence alleles of the TPA-25 insertion were denoted as p and q, respectively (Table 4). Genotype counts for 400/400, 400/100, and 100/100 individuals were denoted as N_{44}, N_{41}, and N_{11}, respectively, and the total sample size was denoted as N (Table 2).

$$p = \frac{2N_{44} + N_{41}}{2N} \quad (1)$$

Using these equations, the frequency of the 400 bp insertion allele was estimated as $p = 0.542$, while the 100 bp allele frequency was $q = 0.458$. Based on these allele frequencies, the expected genotype frequencies under Hardy–Weinberg equilibrium were calculated as p^2 (400/400), $2pq$ (400/100), and q^2 (100/100) and are presented in Table 4.

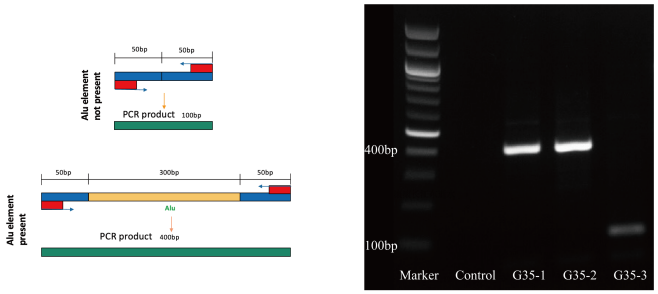
Given a sample size of 191 individuals, the expected genotype counts were 56 individuals with genotype 400/400, 95 with genotype 400/100, and 40 with genotype 100/100 (Figure 1C, Table 2).

4.1.4 Hardy–Weinberg Equilibrium Test

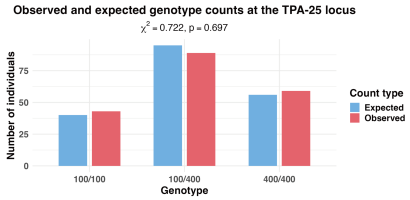
A Pearson chi-square goodness-of-fit test was used to evaluate whether the observed genotype distribution deviated from Hardy–Weinberg expectations. The null hypothesis (H_0) assumed that genotype frequencies follow the Hardy–Weinberg equilibrium based on the estimated allele frequencies. The chi-square statistic was calculated using Equation (3), where O and E represent observed and expected genotype counts, respectively.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad (3)$$

The test yielded $\chi^2 = 0.722$ with two degrees of freedom ($p = 0.697$). This result indicates that the observed genotype distribution does not deviate significantly from the Hardy–Weinberg equilibrium.



(A) (B)



(C)

Figure 1. *PCR genotyping of the TPA-25 locus.* ***(A).*** *Diagram of the TPA-25 Alu insertion and the corresponding PCR fragment sizes (100 bp without insertion; 400 bp with insertion) (from course lecture slides).* ***(B).*** *Gel electrophoresis showing representative genotypes of Group 35 samples.* ***(C).*** *Comparison of observed and expected genotype counts at the TPA-25 locus.*

Sample	***100bp band(s)***	***400bp band(s)***	***Genotype***
G35-1	*Absence*	*Presence*	*400/400*
G35-2	*Absence*	*Presence*	*400/400*
G35-3	*Presence*	*Absence*	*100/100*

Table 2. *Genotype classification of TPA-25 for Group 35 samples.*

	400/400 (N_{44})	***400/100 (N_{41})***	***100/100 (N_{11})***	***Validated (N)***
Observed	*59*	*89*	*43*	*191*
Expected	*56*	*95*	*40*	*191*

Table 3. *Observed and Hardy–Weinberg expected genotype counts for the TPA-25 locus.*
26 samples were excluded due to low-quality gel bands, leaving 191 validated individuals for analysis.

Terms of Frequency		***Frequency***
Allele Frequency	*400 (p)*	*0.542*
	100 (q)	*0.458*
Genotype Frequency	*400/400 (p^2)*	*0.294*
	100/400 (2pq)	*0.496*
	100/100 (q^2)	*0.210*

Table 4. *Allele frequencies and Hardy–Weinberg expected genotype frequencies for the TPA-25 locus.*

4.2 VWA Data Analysis

4.2.1 Gel Treatment

Different experimental groups produced multiple gels. Across all gels, only bands meeting the criteria for reliable fragment analysis were retained, yielding 166 validated sample bands in total. Bands were excluded if they were clearly above 200 bp, showed features consistent with primer dimers or other non-specific amplification products, occurred in lanes with three bands (Figure 2A; Question h), appeared on visibly tilted gel images likely to introduce measurement error, or produced estimated fragment sizes falling clearly outside the expected 138–162 bp range by more than one repeat unit (±4 bp). For clarity, the gel from Group 3–4 was selected as a representative example for the measurement and calibration workflow (Figure 2B), because it displayed a clearly resolved DNA ladder and well-separated bands.

4.2.2 Measurement of Band Migration Distances

Band migration distances were measured from the representative Group 3–4 gel using ImageJ. The distance from the bottom of the well to the center of each band was recorded in pixels for each lane (Table 5). Measurements were obtained for both DNA ladder bands and sample bands.

4.2.3 DNA Ladder Calibration

Migration distances of ladder bands with known fragment sizes were used to construct a calibration curve relating migration distance to fragment size. As the DNA migration distance in gel electrophoresis is approximately proportional to the logarithm of fragment size, a linear regression model was fitted according to Equation (4):

$$log_{10}(bp) = a + b * distance \qquad (4)$$

A strong linear relationship was observed between migration distance and the logarithm of fragment size, with a coefficient of determination of $R^2 = 0.999$ (Figure 2C), demonstrating validation of our calibration.

4.2.4 Fragment Size Prediction and Allele Classification

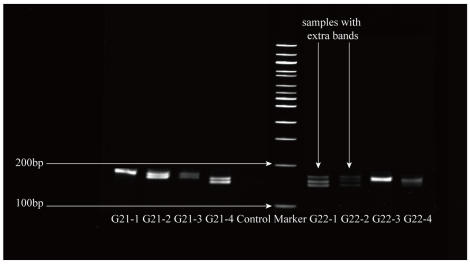
Using the calibration model, migration distances of the Group 3–4 sample bands were converted into estimated fragment sizes, which were rounded to integer values (Figure 2D; Table 5). The same procedure was then applied to all 166 validated bands for distribution analysis. The VWA locus contains tetranucleotide repeat units (4 bp). Consequently, allele sizes differ in multiples of 4 bp. Estimated fragment sizes were therefore assigned to allele classes ranging from 138 bp to 162 bp.

4.2.5 Distribution Pattern Analysis

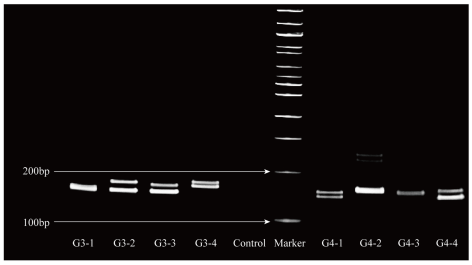
The distribution pattern was then analyzed using all 166 validated bands across groups. All observed allele classes fell within the expected 138–162 bp range, with 138 bp as the smallest class and 162 bp as the largest (Figure 2E). A total of seven allele classes were detected. The 154 bp allele was the most frequent, with 41 observations, whereas the 138 bp allele was the least frequent, with 8 observations.

Across all validated bands, the absolute difference between estimated fragment sizes and the nearest allele class ranged from 0 to 3 bp (e.g., Table 5), which is smaller than one repeat unit (4 bp). Given that VWA alleles vary in 4 bp increments, the measurement uncertainty was therefore estimated as approximately ±1 repeat unit (±4 bp).

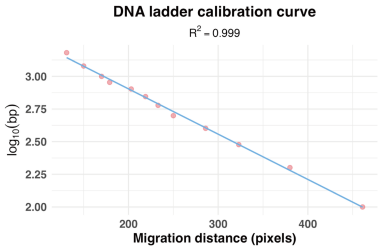
Overall, the observed distribution resembles the bell-shaped unimodal pattern commonly reported for microsatellite loci (12). However, the limited sample size in this dataset does not permit further statistical evaluation of the precise distribution pattern.



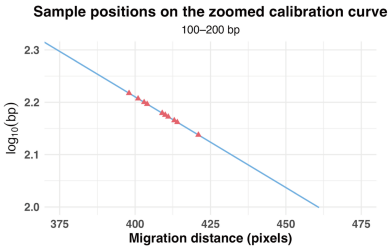
(A)



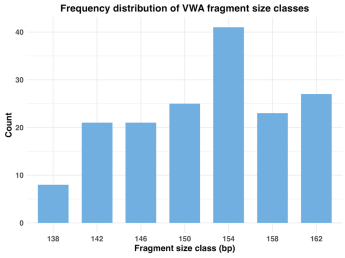
(B)



(C)



(D)



(E)

Figure 2. *Analysis of VWA fragment sizes and allele distribution.* ***(A).*** *Example of extra band patterns observed during VWA genotyping, which were excluded from the analysis.* ***(B).*** *Representative gel electrophoresis image selected from Group 3–4 for the measurement and calibration workflow.* ***(C).*** *DNA ladder calibration curve relating migration distance to* $\log_{10}$*(fragment size).* ***(D).*** *Zoomed view of the calibration curve (100–200 bp) showing the projected positions of the Group 3–4 sample bands.* ***(E).*** *Frequency distribution of VWA fragment size classes based on all 166 validated sample bands.*

Sample	***Migration Distance (pixels)***	***Estimated Size (bp)***	***Size Classification (bp)***	***Classification Error (bp)***
G3-1	*404*	*157*	*158*	*1*
	404	*157*	*158*	*1*
G3-2	*398*	*165*	*162*	*3*
	409	*151*	*150*	*1*
G3-3	*401*	*161*	*162*	*1*
	413	*146*	*146*	*0*
G3-4	*398*	*165*	*162*	*3*
	403	*158*	*158*	*0*
G4-1	*414*	*145*	*146*	*1*

G4-2	*421*	*137*	*138*	*1*
	409	*151*	*150*	*1*
	409	*151*	*150*	*1*
G4-3	*410*	*150*	*150*	*0*
	410	*150*	*150*	*0*
G4-4	*411*	*149*	*150*	*1*
	421	*137*	*138*	*1*

Table 5. *Statistics of Group 3–4 VWA PCR bands. Fragment sizes were predicted using the ladder calibration model and assigned to the nearest allele class (138–162 bp, 4-bp increments). Classification error represents the absolute difference between the estimated fragment size and the assigned allele class.*

5. Conclusion & Discussion

5.1Conclusion

This study examined human genetic variation using two classes of molecular markers: the TPA-25 Alu insertion locus and the VWA microsatellite locus, representing a SINE insertion polymorphism and an STR marker, respectively.

For genotype frequencies at the TPA-25 locus, the observed genotype distribution did not significantly differ from the expected distribution ($\chi^2 = 0.722$, $p = 0.697$), indicating that the analyzed population is consistent with the Hardy–Weinberg equilibrium at this locus. This result supports the HW principle, which predicts stable allele and genotype frequencies across generations in the absence of evolutionary forces such as selection, mutation, migration, or non-random mating, and in sufficiently large populations where genetic drift is minimal. The estimated insertion allele frequency ($p = 0.542$) is also similar to values reported for Asian populations, including Japan ($p = 0.576$) (8), further supporting the reliability of the genotyping results.

The second experiment analyzed the VWA microsatellite locus. Two key polymorphism properties of the VWA locus were consistent with the observations. First, allele sizes ranged from 138 to 162 bp. Second, allele variation occurred in 4 bp increments, reflecting the tetranucleotide repeat structure of the locus. The observed allele distribution also approximated the unimodal pattern commonly reported for microsatellite loci. Furthermore, considerable variation in allele composition was observed among individuals, illustrating the high polymorphism of this marker. Such variability underlies the widespread use of VWA and other microsatellite loci in genetic identification, including forensic analysis and parentage testing.

Overall, this study demonstrates that molecular genotyping, combined with population genetic theory, can reveal patterns of human genetic variation. The analyses also illustrate the utility of polymorphic loci such as TPA-25 and VWA for investigating population structure or evolution, and for applications in genetic identification.

5.2 Discussion

Both experiments present several methodological limitations beyond common issues such as missing repeats or potential batch effects, which should be considered when interpreting the results.

For TPA-25 genotyping, the analyzed individuals were all students from the same practical class, meaning that the sampled population may not represent a randomly sampled population. Consequently, the assumption of random sampling required for the Hardy–Weinberg analysis may not be fully satisfied. Since previous studies have reported that TPA-25 Alu insertion allele frequencies vary across human populations and geographic regions (8), the allele frequency estimated in this study should be

interpreted cautiously and may not accurately reflect the broader population.

Several technical limitations were also identified in the VWA microsatellite analysis. From a statistical perspective, the number of analyzed samples was relatively small, limiting the ability to perform more detailed analyses of allele distribution patterns, such as evaluating distribution skewness or kurtosis. Future studies could improve this limitation by analyzing larger sample sets and incorporating additional metadata, including variables such as age, sex, or geographic origin. Such information would enable further investigation of potential associations between demographic factors and genetic variation at the VWA locus, as well as possible interactions among these variables. Experimental artefacts such as limited resolution of polyacrylamide gel, PCR stutter bands, and heteroduplex formation also affect fragment size estimation. Details are discussed in the following Question h and j.

6. Answers to Questions

6.1 TPA-25 Genotyping

a) Yes, the PCR was successful.
The negative control (H_2O) showed no amplification, indicating the absence of contamination. Moreover, clear bands were observed for both marker and all samples at the expected sizes (100 bp and 400 bp) (Figure 1B, Table 2).

b) Yes, with high confidence.
Principally, the samples are from our cheek cells, which contain our DNA. Also, samples were labelled and processed individually throughout extraction, PCR amplification, and electrophoresis. Moreover, the clean negative control lane further supports the absence of contamination to the amplified DNA.

c) Samples G35-1 and G35-2 are homozygous for the presence of the TPA-25 insertion (400/400), while G35-3 is homozygous for the absence of the insertion (100/100) (Figure 1B, Table 2).

d) Yes.
Only bands with 400bp appear in Sample G35-1 & G35-2, showing the presence of TPA-25 insertion. Only bands with 100bp appear in Sample G35-3, showing the absence of TPA-25 insertion (Figure 1A).

6.2 VWA Genotyping

e) The largest fragment observed for the VWA locus was approximately 162 bp, while the smallest fragment was approximately 138 bp (Figure 2E).

f) Seven distinct fragment sizes were observed (138, 142, 146, 150, 154, 158, and 162 bp) (Figure 2E).

g) The 154 bp allele was the most common, observed 41 times among the 166 analyzed alleles. The 138 bp alleles were the rarest with eight observations (Figure 2E).

h) Except for the possibility of sample contamination, some other explanations for extra fragments (Figure 2A) exist.
One is PCR stutter bands, produced when DNA polymerase undergoes slipped-strand mispairing during STR amplification, generating products differing by one repeat unit (13).
Another possible mechanism is heteroduplex formation. In heterozygous samples, single strands from different alleles may re-anneal during PCR cycles, forming duplexes with mismatched repeat regions. These heteroduplex molecules migrate differently during electrophoresis and may appear as additional bands (14).

i) x ±1 repeat unit (±4 bp)
As we discussed in Section 4.2.4, during classification, the difference between measured fragment sizes and allele classes ranged from 0 to 3 bp, which is smaller than one repeat unit (4 bp) (Table 5).

j) Several approaches could improve experimental accuracy.

First, higher-resolution electrophoresis methods, such as capillary electrophoresis, could provide more precise fragment size determination (5). Increasing the resolution of gel imaging would also improve measurement accuracy.

Second, the calibration DNA ladder could be optimized. Because the analyzed fragments lie within the 138–162 bp range, a ladder with denser size markers in this region would improve accuracy.

Third, PCR optimization may reduce amplification artefacts. Adjusting primer specificity, annealing temperature, or reaction conditions could reduce stutter formation and heteroduplex products.

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